

Word Processing Assignment: Analysis of the Natural Stories Corpus

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February 17, 2026

GitHub Repository: <https://github.com/Ishaank1805/psycholinguistic-reading-time-analysis>

1 Part I: Preliminary Data Analysis

1.1 Dataset and Preprocessing

We used the Natural Stories corpus (Futrell et al., 2021), consisting of 10 stories with self-paced reading time (RT) data from 181 native English speakers collected via Amazon Mechanical Turk. Following the preprocessing described in the original paper, participant-story passes with fewer than 5 correct comprehension answers were excluded, along with individual RTs below 100 ms or above 3000 ms. Word frequencies were obtained from the Google Books N-grams corpus (summing over years 1990–2013), provided in the `freqs-1.tsv` file. After computing mean RT per word position (averaged across subjects) and merging with frequency data, 10,246 unique word positions were available for analysis.

1.2 Q1: Mean RT per Word

Mean RT per word was computed by averaging reading times across all subjects for each unique word position, identified by the pair (item, zone). The corpus-wide average mean RT was approximately 340 ms.

1.3 Q2: Word Length vs Mean RT

Figure 1 shows the relationship between word length (in characters) and mean RT. The scatter plot reveals a positive trend, and the aggregated plot confirms a steady increase in mean RT from approximately 280 ms for 1–2 character words to around 420 ms for 13+ character words. Longer words require more visual processing and more time for lexical access.

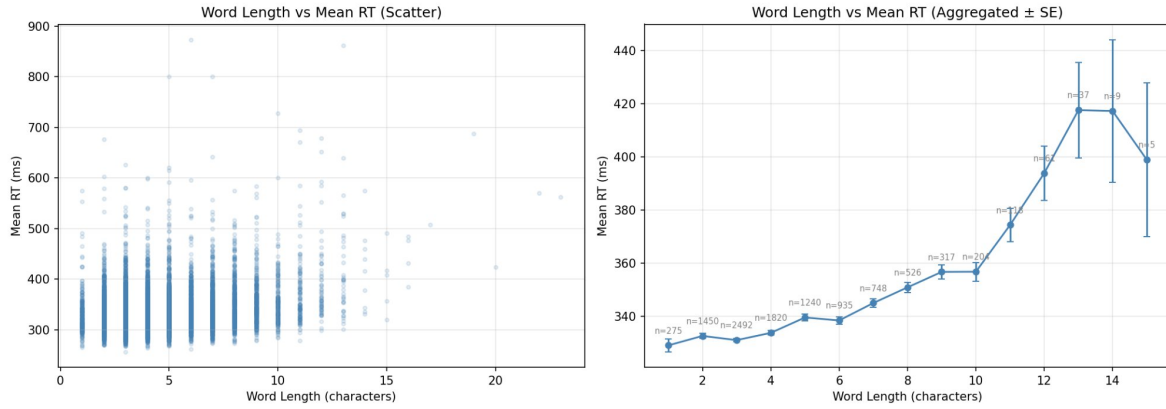


Figure 1: Word length vs mean reading time: scatter plot (left) and aggregated means with standard error bars (right).

1.4 Q3: Word Frequency vs Mean RT

Figure 2 shows the relationship between log-transformed word frequency and mean RT. The binned plot demonstrates a clear downward trend: words with the lowest frequencies ($\log_{10}$ frequency ≈ 0) have mean RTs around 550 ms, while the most frequent words ($\log_{10}$ frequency ≈ 10) are read in approximately 320 ms.

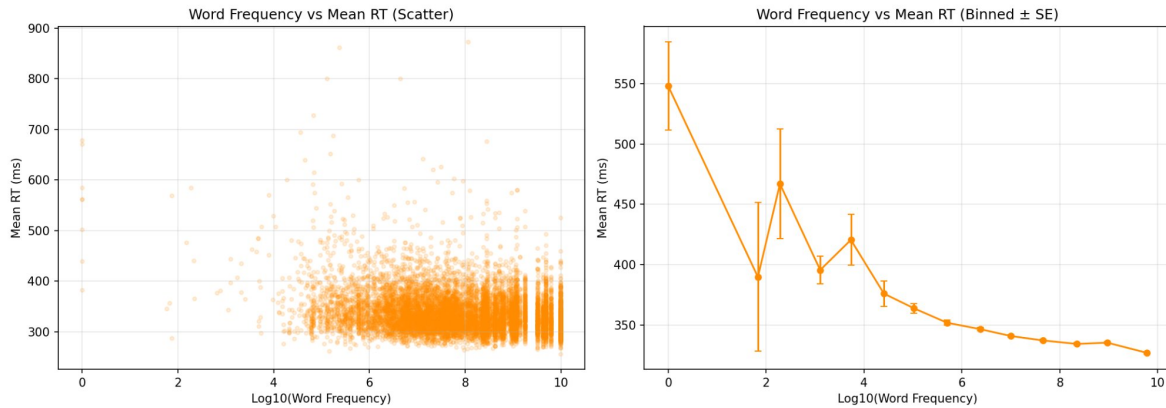


Figure 2: Word frequency vs mean reading time: scatter plot (left) and binned means with standard error bars (right).

1.5 Q4–Q6: Pearson Correlations

Table 1 summarises the Pearson correlation coefficients. All correlations are highly significant ($p \approx 0$).

Table 1: Pearson correlation coefficients between word length, log frequency, and mean RT.

Variable Pair	r	p-value
Word Length vs Log Frequency	-0.7202	≈ 0
Word Length vs Mean RT	+0.2462	2.61×10^{-141}
Log Frequency vs Mean RT	-0.2282	4.23×10^{-121}

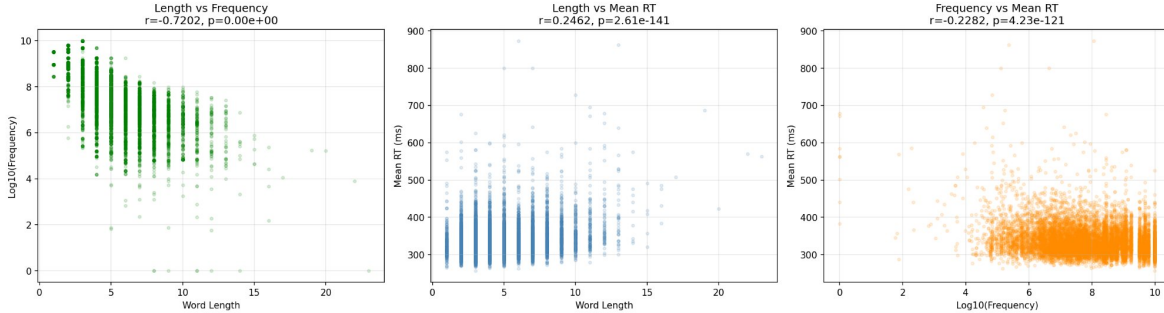


Figure 3: Scatter plots with Pearson correlations for all three variable pairs.

1.6 Summary

A strong negative correlation ($r = -0.72$) exists between word length and log frequency, consistent with Zipf’s Law of Abbreviation: shorter words tend to be more frequent. Word length is positively correlated with mean RT ($r = 0.25$), indicating that longer words take more time to read. Log frequency is negatively correlated with mean RT ($r = -0.23$), confirming that more frequent words are read faster. These results replicate the basic psycholinguistic effects reported by Futrell et al. (2021) in Table 3 of the reference paper: a negative coefficient for log frequency and a positive coefficient for word length. The strong confound between length and frequency motivates the multiple regression analyses in Part II.

2 Part II: Hypothesis Testing

GPT-3 (davinci) surprisal values were obtained from the `all_stories_gpt3.csv` file provided with the corpus. Since GPT-3 tokenises punctuation separately from words, tokens were aligned to RT words by accumulating GPT-3 tokens until they reconstructed each RT word, summing log-probabilities. Surprisal was computed as $-\log_2 P(\text{word} \mid \text{context})$. Content and function words were classified using a comprehensive list of approximately 200 English function words (determiners, pronouns, prepositions, conjunctions, auxiliaries). All remaining words were classified as content words.

2.1 Hypothesis 1: LM Probabilities vs Word Frequency

Hypothesis: Language model probabilities are better predictors of reading time than word frequency.

Table 2: Hypothesis 1: Model comparison.

	Formula	R^2
Model 1	Mean RT $\sim$ log_freq + word_length	0.0658
Model 2	Mean RT $\sim$ GPT-3 surprisal + word_length	0.0775

Model 2 (GPT-3 surprisal + word length) provides a better fit than Model 1 (word frequency + word length), with an R^2 of 0.0775 compared to 0.0658. This supports the hypothesis that language model probabilities are better predictors of reading time than word frequency alone.

The advantage of GPT-3 surprisal likely arises because it captures contextual predictability—how expected a word is given the preceding sentence context—rather than just how common a word is in isolation. Both models explain a relatively small proportion of total variance ($< 8\%$), indicating that many other factors (syntactic complexity, semantic plausibility, individual differences) also contribute to reading time variation.

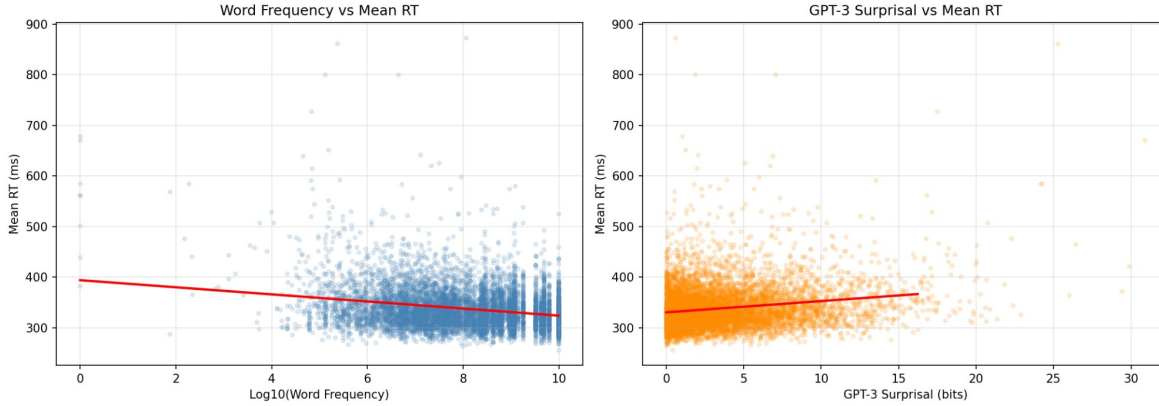


Figure 4: Scatter plots of word frequency (left) and GPT-3 surprisal (right) against mean RT, with regression lines.

Hypothesis 1: Model Comparison

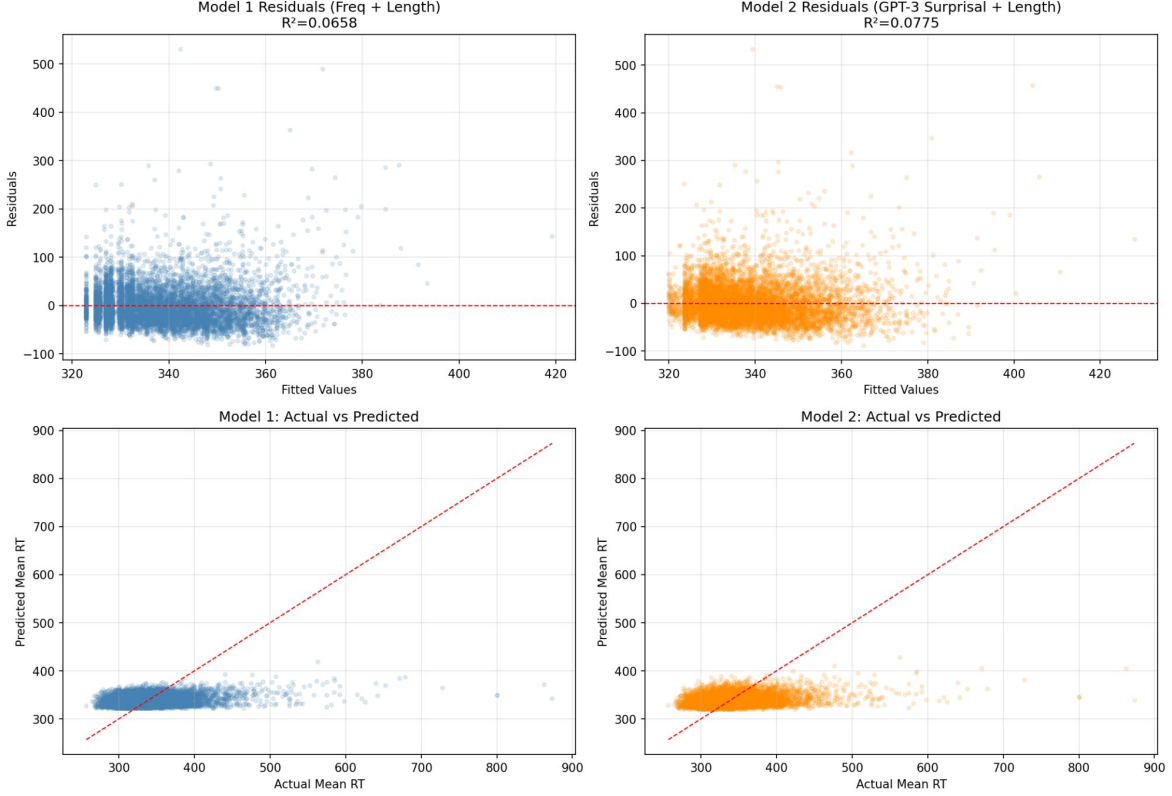


Figure 5: Hypothesis 1: Residual plots (top) and actual vs predicted plots (bottom) for both models. R^2 values are shown in the subplot titles.

2.2 Hypothesis 2: Content Words vs Function Words

Hypothesis: Content words are processed differently than function words.

Table 3: Hypothesis 2: Model comparison across word types.

Model	Word Type	Predictor	R^2	AIC
M1	Content	Freq + Length	0.1028	40783
M2	Content	Surprisal + Length	0.0991	40799
M3	Function	Freq + Length	0.0230	40557
M4	Function	Surprisal + Length	0.0306	40525

The results confirm that content words are processed differently from function words. Content word models explain substantially more variance ($R^2 \approx 0.10$) than function word models ($R^2 \approx 0.02$ – 0.03). For content words, frequency slightly outperforms surprisal ($R^2 = 0.1028$ vs 0.0991), suggesting that raw lexical frequency is a marginally better predictor for open-class words. For function words, surprisal slightly outperforms frequency ($R^2 = 0.0306$ vs 0.0230). Since function words are predominantly high-frequency and short, there is little

variance in frequency to explain RT differences; GPT-3 surprisal captures contextual factors that create the small observed RT variations.

The distribution plots (Figure 7) confirm fundamental differences: function words are concentrated at high frequencies ($\log_{10} \text{freq} > 7$) and low surprisal (< 5 bits), while content words span a much wider range of both measures. These results are consistent with dual-route models of lexical access.

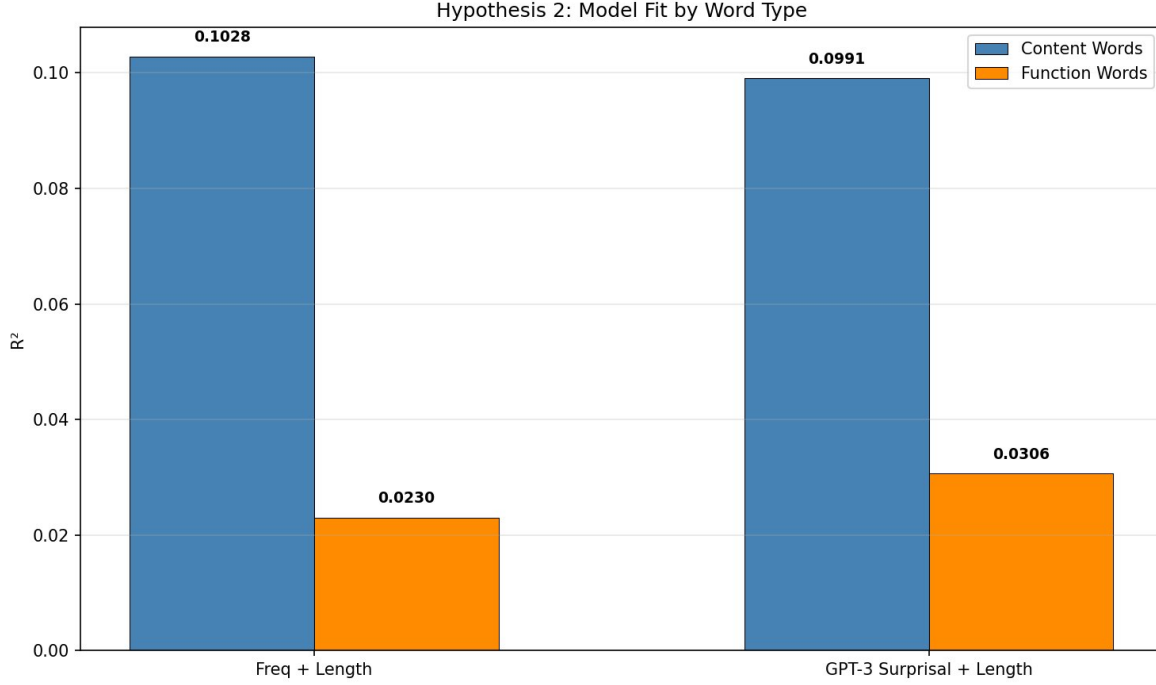


Figure 6: Hypothesis 2: R^2 comparison across word types and predictor models.

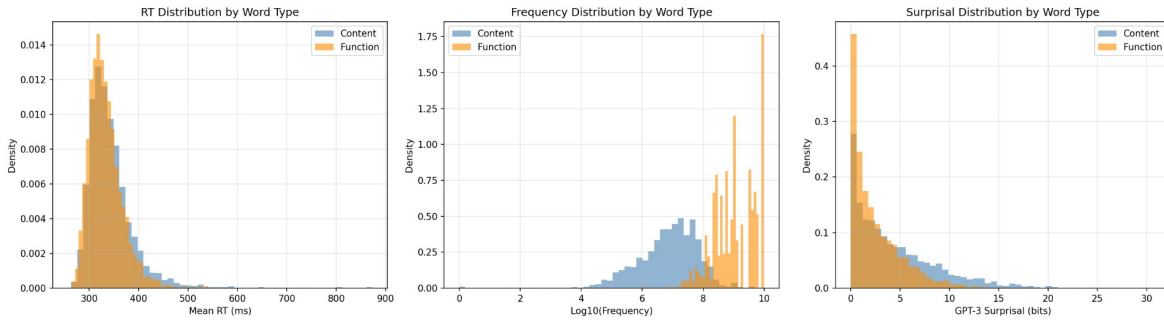


Figure 7: Distributions of RT, frequency, and surprisal for content vs function words.

Hypothesis 2: Predictors vs RT by Word Type

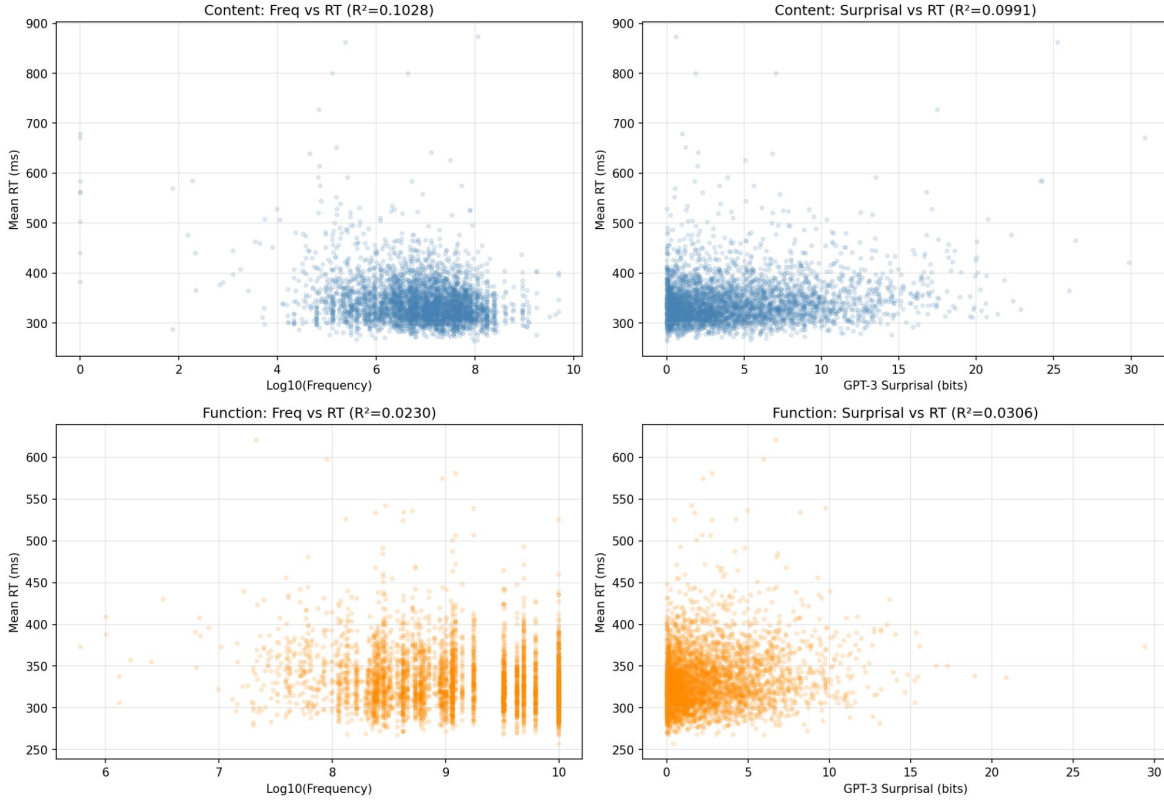


Figure 8: Predictors vs RT scatter plots separated by word type, with R^2 values.

3 Part III: Frequency Ordered Bin Search (FOBS)

3.1 FOBS Model Construction

We constructed a FOBS model by lemmatising all words using the NLTK WordNet lemmatiser with POS tagging. Lemma frequencies were computed by summing the Google Books corpus frequencies of all surface forms sharing the same root. Lemmas were sorted by frequency and assigned to 10 frequency-ordered bins (Bin 1 = most frequent, Bin 10 = least frequent).

Figure 9 shows that mean RT increases monotonically from approximately 325 ms in Bin 1 to 371 ms in Bin 10, confirming the core FOBS prediction: lexical access time increases as the search proceeds through lower-frequency bins.

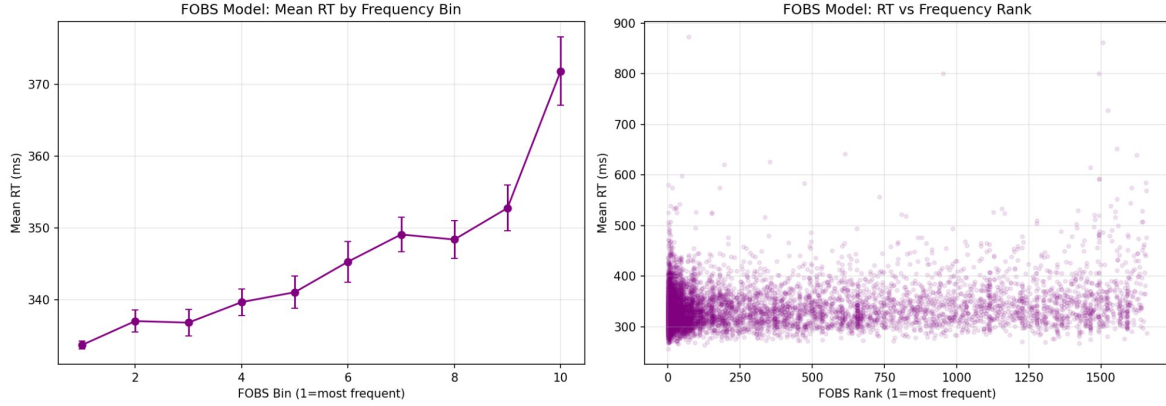


Figure 9: FOBS model: Mean RT by frequency bin (left) and RT vs frequency rank (right).

3.2 Hypothesis 1: Root Frequency vs Surface Frequency

Hypothesis: Root frequency predicts reading times better than surface frequency.

Table 4: FOBS Hypothesis 1: Surface vs lemma frequency models.

	Formula	R^2	AIC
Model 1	Mean RT $\sim$ surface_freq + word_length	0.0576	81530
Model 2	Mean RT $\sim$ lemma_freq + lemma_length	0.0483	81608

Contrary to the hypothesis, surface frequency is a slightly better predictor of reading time than root frequency ($R^2 = 0.0576$ vs 0.0483 ; AIC = 81530 vs 81608). This suggests that readers access words primarily via their surface forms rather than through obligatory morphological decomposition. However, the difference is modest (~ 1 percentage point), indicating that both measures capture similar variance—which is expected given the high correlation between a word’s frequency and its lemma’s frequency.

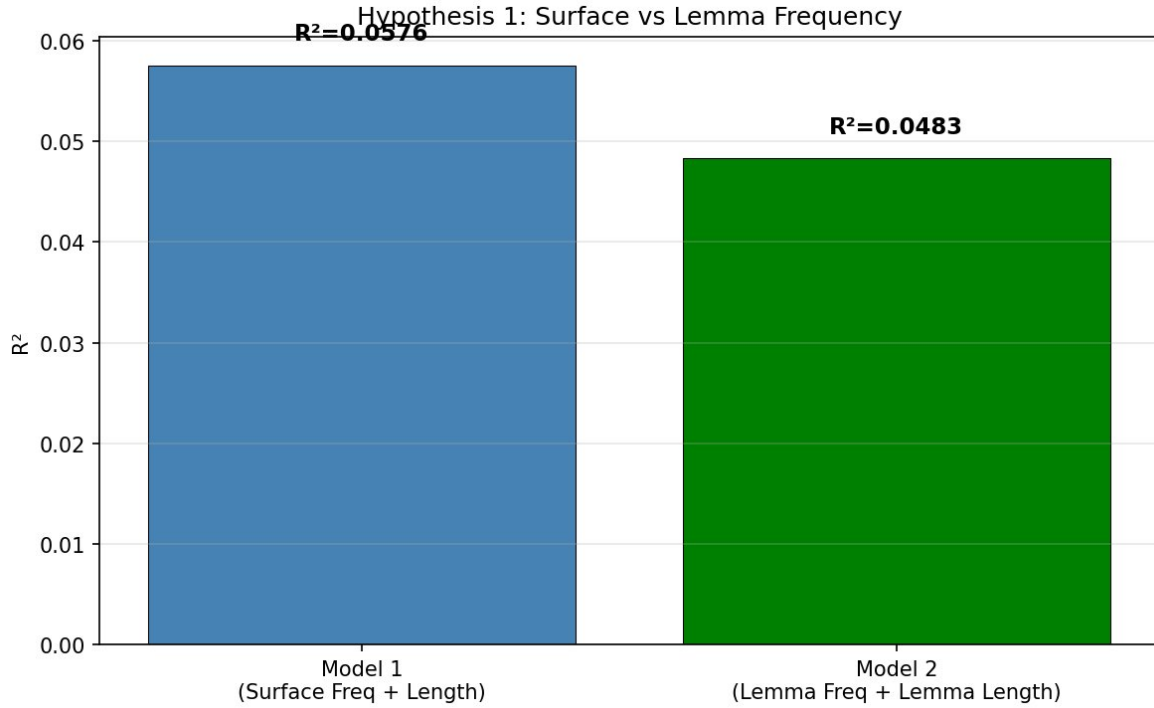


Figure 10: FOBS Hypothesis 1: R^2 comparison of surface vs lemma frequency models.

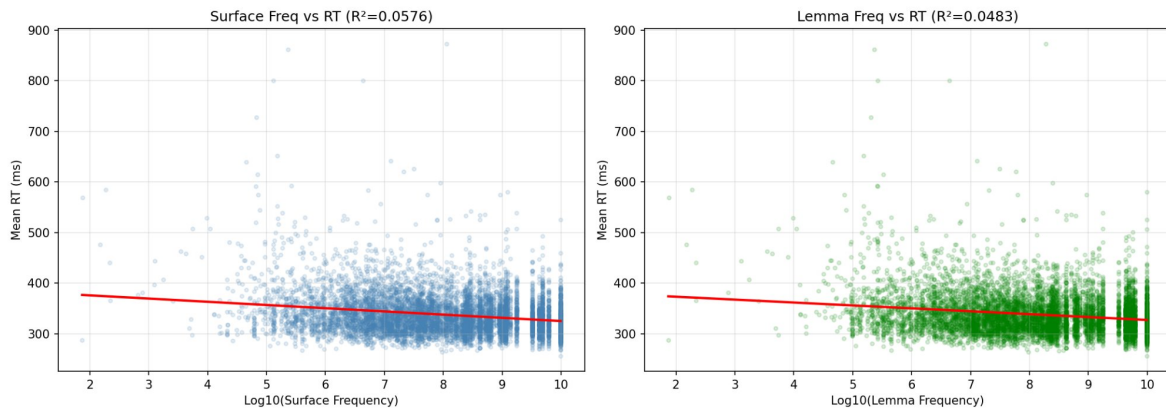


Figure 11: Surface frequency (left) and lemma frequency (right) vs mean RT with regression lines.

3.3 Hypothesis 2: Pseudo-Affixed vs Regularly Affixed Words

Hypothesis: Pseudo-affixed words like “finger” take more processing time compared to words with regular affixes like “driver”.

We identified words of equal length (6 characters) in the corpus ending in *-er* and classified them as pseudo-affixed (the apparent root does not exist or is semantically unrelated) or

regularly affixed (the root is a real word with a transparent derivational relationship). Five words were selected per group, matched on word length and approximate frequency.

Table 5: Pseudo-affixed vs regularly affixed word comparison (all 6 characters).

Group	Word	Log Freq	Mean RT (ms)
Pseudo-affixed	finger	6.71	402.4
	corner	6.95	316.2
	master	7.03	329.7
	silver	6.81	396.1
	sister	7.00	336.4
Mean		6.90	356.2
Regular-affixed	closer	6.92	342.2
	higher	7.32	340.9
	larger	7.28	343.5
	longer	7.45	321.1
	lesser	6.49	350.5
Mean		7.09	339.6

Pseudo-affixed words showed a numerically higher mean RT (356.2 ms vs 339.6 ms), consistent with the hypothesis that failed morphological decomposition incurs an additional processing cost. However, the difference was not statistically significant ($t = 0.89$, $p = 0.400$). This null result is likely attributable to the small sample size (5 words per group), high within-group variance (e.g., “finger” at 402 ms vs “corner” at 316 ms), and the limited sensitivity of self-paced reading to small morphological processing effects. Primed lexical decision tasks, which isolate lexical access more precisely, would be better suited to test this hypothesis.

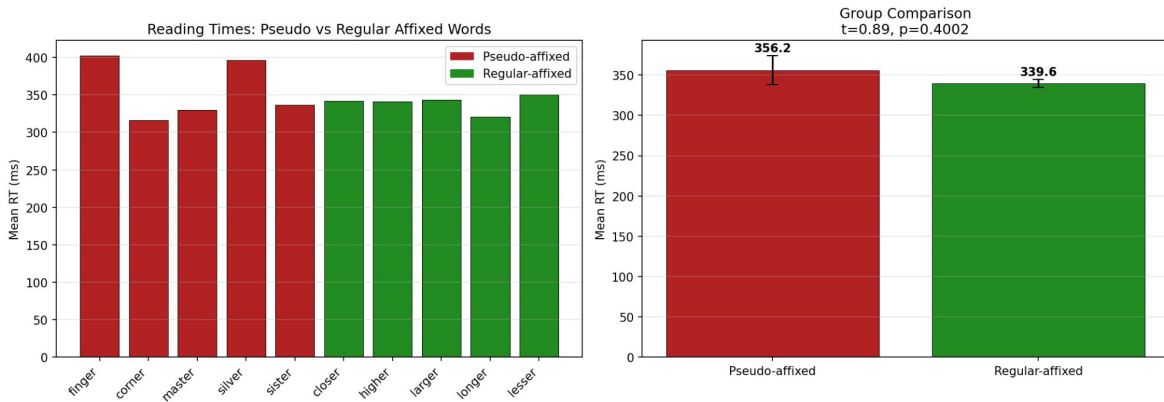


Figure 12: Reading times for individual pseudo- and regularly affixed words (left) and group means with standard error (right).

References

- Futrell, R., Gibson, E., Tily, H. J., Blank, I., Vishnevetsky, A., Piantadosi, S. T., & Fedorenko, E. (2021). The Natural Stories corpus: a reading-time corpus of English texts containing rare syntactic constructions. *Language Resources and Evaluation*, 55, 63–77. <https://doi.org/10.1007/s10579-020-09503-7>
- Natural Stories Corpus Repository: <https://github.com/languageMIT/naturalstories>
- Traxler, M. J. (2012). *Introduction to Psycholinguistics: Understanding Language Science*. Wiley-Blackwell. Chapter 3.